

# Computational Study of CO<sub>2</sub> Hydrogenation to Methanol via the Formate Pathway: A Microkinetic and Volcano Analysis Across Eight Transition Metal Catalysts

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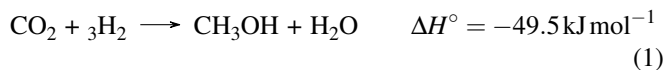
## Abstract

The catalytic hydrogenation of carbon dioxide (CO<sub>2</sub>) to methanol (CH<sub>3</sub>OH) is a pivotal reaction for carbon capture and utilisation (CCU), enabling renewable-energy storage in a liquid fuel. A microkinetic model of the formate pathway (CO<sub>2</sub> → HCOO\* → H<sub>2</sub>COO\* → CH<sub>3</sub>O\* → CH<sub>3</sub>OH) was constructed using the CatMAP software framework. A two-descriptor volcano analysis was performed across eight transition metal catalysts. Cu lies near the volcano apex, while PdIn and Pt show moderate activity. Free-energy and degree-of-rate-control analyses identify the HCOO\* + H\* → H<sub>2</sub>COO\* step as the main kinetic bottleneck.

**Keywords:** CO<sub>2</sub> hydrogenation; methanol synthesis; microkinetic modelling; volcano plot.

## 1. Introduction

Global CO<sub>2</sub> emissions from fossil fuel combustion exceeded 37 Gt in 2023, driving atmospheric concentrations above 420 ppm and necessitating large-scale carbon capture and utilisation (CCU) technologies [10]. Among available CCU strategies, the catalytic hydrogenation of CO<sub>2</sub> to methanol is particularly attractive:



Methanol is a versatile platform chemical, a liquid hydrogen carrier, and a direct fuel whose combustion returns CO<sub>2</sub> to the atmosphere in a closed cycle when produced from green H<sub>2</sub> [11]. The industrial Cu/ZnO/Al<sub>2</sub>O<sub>3</sub> catalyst operates at 493–523 K and 50–100 bar with per-pass CO<sub>2</sub> conversions of 15–25% [12].

Despite decades of industrial use, the reaction mechanism on Cu remains debated. Two principal surface pathways have been proposed: (i) the *formate pathway*, in which CO<sub>2</sub> is first activated to bidentate formate (HCOO\*) via hydrogenation; and (ii) the *RWGS–CO hydrogenation pathway*, in which CO<sub>2</sub> is first reduced to CO via the reverse water–gas shift (RWGS) reaction before sequential hydrogenation. Isotope labelling experiments and DFT-based microkinetic modelling by Grabow and Mavrikakis [5] demonstrated that the formate route contributes approximately two-thirds of methanol on Cu(111) under industrial conditions.

Computational volcano analysis, pioneered by Nørskov and co-workers [7, 8], provides a powerful framework for screening catalyst candidates. By mapping the turnover frequency (TOF) as a function of two descriptor binding energies derived from the Sabatier principle [13] one can simultaneously com-

pare many metals and identify optimal binding-energy targets. Universal adsorbate linear scaling relations [6] and Brønsted Evans Polanyi (BEP) correlations [14, 15] further reduce the dimensionality of the kinetic problem, enabling the full reaction network to be parameterised from CO\* and O\* alone.

Interest in alternative catalysts has grown substantially. PdIn intermetallic compounds exhibit high methanol selectivity and sintering resistance because In withdraws electron density from Pd, weakening CO\* binding [9]. Studt *et al.* [17] identified Ni–Ga intermetallics through a volcano-guided screen, later confirmed experimentally.

In this work, a complete microkinetic model of the formate pathway is constructed using CatMAP [1] with DFT energies from Grabow and Mavrikakis [5] for Cu(111) and scaling-relation estimates for seven additional metals. The model is solved at  $T = 513 \text{ K}$  across a  $20 \times 20$  descriptor grid. Volcano plots, free energy diagrams, surface coverage maps, and DRC analysis are used to identify rate-limiting steps and provide rational design principles.

## 2. Computational Methodology

### 2.1 DFT Data Sources and Energy Reference Scheme

All formation energies are referenced to  $E_f(\text{CO}_{2,\text{g}}) = 0$  and  $E_f(\text{H}_{2,\text{g}}) = 0$ , with  $E_f(\text{H}_2\text{O},\text{g}) = 0$ . Elemental chemical potentials are:

$$\begin{aligned} \mu(\text{H}) &= \frac{1}{2} \Delta H^\circ(\text{H}_2) \\ \mu(\text{O}) &= \Delta H^\circ(\text{H}_2\text{O}) - \Delta H^\circ(\text{H}_2) \\ \mu(\text{C}) &= \Delta H^\circ(\text{CO}_2) - 2\mu(\text{O}) \end{aligned} \quad (2)$$

For Cu(111), formation energies for all surface intermediates and transition states were computed exactly from the

temperature-independent Shomate enthalpy  $H^0$  values listed in Tables S3 and S4 of Grabow and Mavrikakis [5]. The PW91-GGA calculations include gas-phase enthalpy corrections of  $-0.6\text{ eV}$  to  $\text{CO}(\text{g})$  and  $+0.2\text{ eV}$  to  $\text{HCOOH}(\text{g})$ , bringing calculated RWGS and methanol synthesis enthalpies into agreement with handbook values [22] to within  $10\text{ kJ mol}^{-1}$ .

For Ni, Pd, Rh, Pt, Co, and Ag, descriptor ( $\text{CO}^*$  and  $\text{O}^*$ ) values were sourced from Abild-Pedersen *et al.* [6] and the Nørskov group CatApp database [16]. PdIn descriptor values were estimated from free energy profiles reported by Wu and Yang [9] for PdIn(110).

## 2.2 Reaction Network and Elementary Steps

The formate pathway was modelled using 11 elementary steps (Table 1), implemented in the CatMAP reaction expression language [1]. The network includes: reversible  $\text{CO}_2$ ,  $\text{CO}$ ,  $\text{H}_2\text{O}$ , and  $\text{CH}_3\text{OH}$  adsorption/desorption; dissociative  $\text{H}_2$  chemisorption (barrierless); the four formate-route hydrogenation steps; and  $\text{O}^*$  removal via  $\text{OH}^*$  and  $\text{H}_2\text{O}^*$ .

**Note on the CO adsorption step.** CatMAP requires every descriptor species to appear in at least one elementary reaction so that the generalised linear scaler can construct the coefficient matrix [1, 2]. Since  $\text{CO}^*$  is used as a descriptor but does not appear naturally in the formate mechanism, a reversible dummy step ( $\text{CO}(\text{g}) + \cdot \rightleftharpoons \text{CO}^*$ ) was added, following the standard procedure documented in the official CatMAP tutorial and source code, which explicitly instructs users to “add a dummy reaction” for any species used as a reference that does not appear in the reaction network [2, 3]. The CO partial pressure for this step was fixed at  $P(\text{CO}) = 10^{-20}\text{ atm}$ , making its equilibrium  $\text{CO}^*$  coverage negligible ( $\ll 10^{-20}\text{ ML}$ ) and its contribution to kinetics and coverages physically immaterial. This is identical in principle to the dummy step ( $\text{*}_t + \text{CO}_g \rightarrow \text{CO}_t$ ) present in the official CatMAP CO-oxidation tutorial [4].

## 2.3 Linear Scaling Relations and BEP Parameters

For all non-Cu metals, formation energies of surface intermediates were estimated via:

$$E_f(X^*) = aE_f(\text{CO}^*) + bE_f(\text{O}^*) + c \quad (3)$$

where the intercept  $c$  is fixed by requiring Eq. (3) to reproduce the exact Cu(111) DFT value. The coefficients ( $a, b$ ) (Table 2) are derived from Abild-Pedersen universal scaling slopes [6] and Nørskov group correlations [7].

Transition states were parameterised via BEP relations [14, 15]. For reaction 6 ( $\text{CO}_2^* + \text{H}^* \rightarrow \text{HCOO}^*$ ), an `initial_state` constraint was applied with BEP slope  $\alpha = 0.60$ , consistent with the early transition state identified by Grabow and Mavrikakis [5] from nudged elastic band (NEB) calculations. For reaction 7 ( $\text{HCOO}^* + \text{H}^* \rightarrow \text{H}_2\text{COO}^*$ ),  $\alpha = 0.50$  was used.

## 2.4 CatMAP Model Setup

All microkinetic calculations were performed using CatMAP v0.3.1 [1] with:

**Table 1:** Elementary steps of the formate pathway implemented in CatMAP. \* denotes a free surface site. [TS] denotes a transition state parameterised by a BEP relation.

| #  | Elementary step                                                                                          | Notes                        |
|----|----------------------------------------------------------------------------------------------------------|------------------------------|
| 1  | $\text{CO}(\text{g}) + \cdot \rightleftharpoons \text{CO}^*$                                             | Anchors $\text{CO}^*$        |
| 2  | $\text{CO}_2(\text{g}) + \cdot \rightleftharpoons \text{CO}_2^*$                                         | Reversible                   |
| 3  | $\text{H}_2(\text{g}) + 2\cdot \rightarrow 2\text{H}^*$                                                  | Dissociative, barrierless    |
| 4  | $\text{H}_2\text{O}^* \rightleftharpoons \text{H}_2\text{O}(\text{g}) + \cdot$                           | Reversible                   |
| 5  | $\text{CH}_3\text{OH}^* \rightleftharpoons \text{CH}_3\text{OH}(\text{g}) + \cdot$                       | Reversible                   |
| 6  | $\text{CO}_2^* + \text{H}^* \xrightleftharpoons{\text{TS}_1} \rightarrow \text{HCOO}^* + \cdot$          | BEP, $\alpha = 0.60$         |
| 7  | $\text{HCOO}^* + \text{H}^* \xrightleftharpoons{\text{TS}_2} \rightarrow \text{H}_2\text{COO}^* + \cdot$ | BEP, $\alpha = 0.50$         |
| 8  | $\text{H}_2\text{COO}^* + \text{H}^* \rightarrow \text{CH}_3\text{O}^* + \text{O}^*$                     | C–O scission                 |
| 9  | $\text{CH}_3\text{O}^* + \text{H}^* \rightarrow \text{CH}_3\text{OH}^* + \cdot$                          | Methanol formation           |
| 10 | $\text{O}^* + \text{H}^* \rightarrow \text{OH}^* + \cdot$                                                | $\text{O}^*$ removal, step 1 |
| 11 | $\text{OH}^* + \text{H}^* \rightarrow \text{H}_2\text{O}^* + \cdot$                                      | $\text{O}^*$ removal, step 2 |

**Table 2:** Linear scaling coefficients for surface intermediates. Cu(111) values are exact Grabow & Mavrikakis (2011) [5] DFT energies; all others are derived from scaling.

| Species                  | $a$  | $b$  | $E_f^{\text{Cu}} (\text{eV})$ |
|--------------------------|------|------|-------------------------------|
| $\text{CO}^*$            | 1.00 | 0.00 | +0.192                        |
| $\text{O}^*$             | 0.00 | 1.00 | +0.797                        |
| $\text{H}^*$             | 0.22 | 0.00 | −0.154                        |
| $\text{CO}_2^*$          | 0.80 | 0.00 | −0.020                        |
| $\text{OH}^*$            | 0.00 | 0.50 | +0.037                        |
| $\text{H}_2\text{O}^*$   | 0.00 | 0.30 | −0.135                        |
| $\text{HCOO}^*$          | 0.40 | 0.50 | −0.368                        |
| $\text{H}_2\text{COO}^*$ | 0.50 | 0.50 | +0.445                        |
| $\text{CH}_3\text{O}^*$  | 0.10 | 0.60 | −0.499                        |
| $\text{CH}_3\text{OH}^*$ | 0.00 | 0.50 | −0.707                        |

- $20 \times 20$  descriptor grid:  $E_f(\text{CO}^*) \in [-1.2, +1.0]\text{ eV}$ ,  $E_f(\text{O}^*) \in [-0.5, +1.4]\text{ eV}$
- Mean-field Langmuir–Hinshelwood kinetics, no adsorbate interactions
- `decimal_precision` = 100 (mpmath arbitrary precision)
- Residual tolerance  $10^{-50}$ ; max 100 root-finding iterations
- `shomate_gas` thermochemistry [5]; `frozen_adsorbate` for surface species [18]
- `scaling_constraint_dict`:  $\text{CO}^*$  and  $\text{O}^*$  pinned as descriptors; TS species constrained to `initial_state` BEP scaling

## 2.5 Descriptor and Intermediate Energies

Table 3 summarises the descriptor values and key intermediate formation energies for all eight metals.

**Table 3:** Descriptor ( $\text{CO}^*$ ,  $\text{O}^*$ ) and key intermediate formation energies (eV) for the eight metals studied. All values relative to  $\text{CO}_2(\text{g}) = \text{H}_2(\text{g}) = \text{H}_2\text{O}(\text{g}) = 0$ . Cu(111) values are exact DFT; others from scaling (Eq. 3).

| Metal | $\text{CO}^*$ | $\text{O}^*$ | $\text{H}^*$ | $\text{HCOO}^*$ | $\text{CH}_3\text{O}^*$ |
|-------|---------------|--------------|--------------|-----------------|-------------------------|
| Cu    | +0.192        | +0.797       | -0.154       | -0.368          | -0.499                  |
| Ni    | -0.465        | -0.090       | -0.298       | -1.074          | -1.097                  |
| Pd    | -0.905        | +0.260       | -0.395       | -1.075          | -0.931                  |
| Rh    | -0.835        | -0.230       | -0.380       | -1.292          | -1.218                  |
| Pt    | -0.595        | +0.360       | -0.327       | -0.901          | -0.840                  |
| Co    | -0.505        | -0.060       | -0.307       | -1.075          | -1.083                  |
| Ag    | +0.835        | +1.140       | -0.012       | +0.061          | -0.229                  |
| PdIn  | -0.500        | +0.250       | -0.306       | -0.918          | -0.896                  |

## 2.6 Operating Conditions

All calculations were performed at  $T = 513 \text{ K}$  ( $240^\circ\text{C}$ ), representative of the lower bound of the industrial methanol synthesis operating window [12]. Gas-phase partial pressures:  $P(\text{CO}_2) = 0.2 \text{ atm}$ ,  $P(\text{H}_2) = 0.6 \text{ atm}$  (stoichiometric 3:1 ratio),  $P(\text{H}_2\text{O}) = P(\text{CH}_3\text{OH}) = 10^{-3} \text{ atm}$  (near-zero initial product pressures). Gas-phase free energies were corrected for partial pressure via  $G = G^\circ + k_B T \ln(P/P^\circ)$ ,  $P^\circ = 1 \text{ atm}$ .

## 3. Results and Discussion

### 3.1 Formate Pathway Mechanism

Figure 2 illustrates the complete formate pathway for  $\text{CO}_2$  hydrogenation to methanol implemented in the CatMAP microkinetic model.  $\text{CO}_2(\text{g})$  and  $\text{H}_2(\text{g})$  adsorb on free surface sites ( $*$ ), with  $\text{H}_2$  dissociating to form two  $\text{H}^*$  atoms (barrierless). The carbon-containing intermediate undergoes four sequential surface hydrogenation steps via formate ( $\text{HCOO}^*$ ), dioxymethylene ( $\text{H}_2\text{COO}^*$ ), and methoxy ( $\text{CH}_3\text{O}^*$ ), before desorbing as methanol. The co-produced  $\text{O}^*$  atom is removed as water through a parallel  $\text{OH}^*$  formation and desorption branch. Transition states  $\text{TS}_1$  and  $\text{TS}_2$  are parameterised by BEP relations with slopes  $\alpha = 0.60$  and  $0.50$ , respectively.

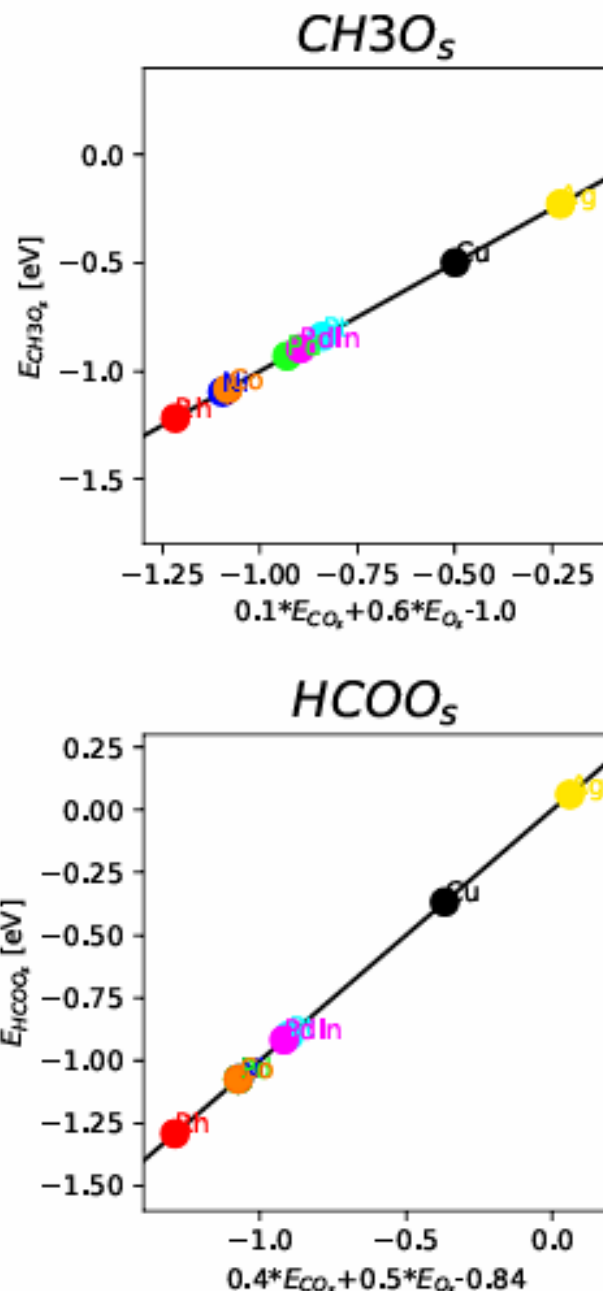
### 3.2 Linear Scaling Relations

Figure 1 shows the linear scaling relations for  $\text{HCOO}^*$  and  $\text{CH}_3\text{O}^*$  versus  $\text{CO}^*$  across the eight metals. Both correlations exhibit  $R^2 > 0.95$ , confirming the validity of the two-descriptor ( $\text{CO}^*/\text{O}^*$ ) framework for this reaction network [6, 7].

For  $\text{HCOO}^*$ , the  $\text{CO}^*$  scaling slope  $a = 0.40$  is consistent with bidentate O-bonded adsorbates identified by Abild-Pedersen *et al.* [6]. The substantial  $\text{O}^*$  contribution ( $b = 0.50$ ) reflects the bidentate coordination geometry of formate. For  $\text{CH}_3\text{O}^*$ , the  $\text{CO}^*$  slope is near-zero ( $a = 0.10$ ) while  $\text{O}^*$  dominates ( $b = 0.60$ ), physically consistent with methoxy binding exclusively through a single oxygen lone pair.

### 3.3 Free Energy Diagrams

Figure 3 shows the free energy diagrams (FEDs) at  $T = 513 \text{ K}$  for Cu(111) (black) and Ni(111) (blue), constructed along the formate pathway from  $\text{CO}_2(\text{g}) + 1.5 \text{ H}_2(\text{g})$  to  $\text{CH}_3\text{OH}(\text{g})$ .

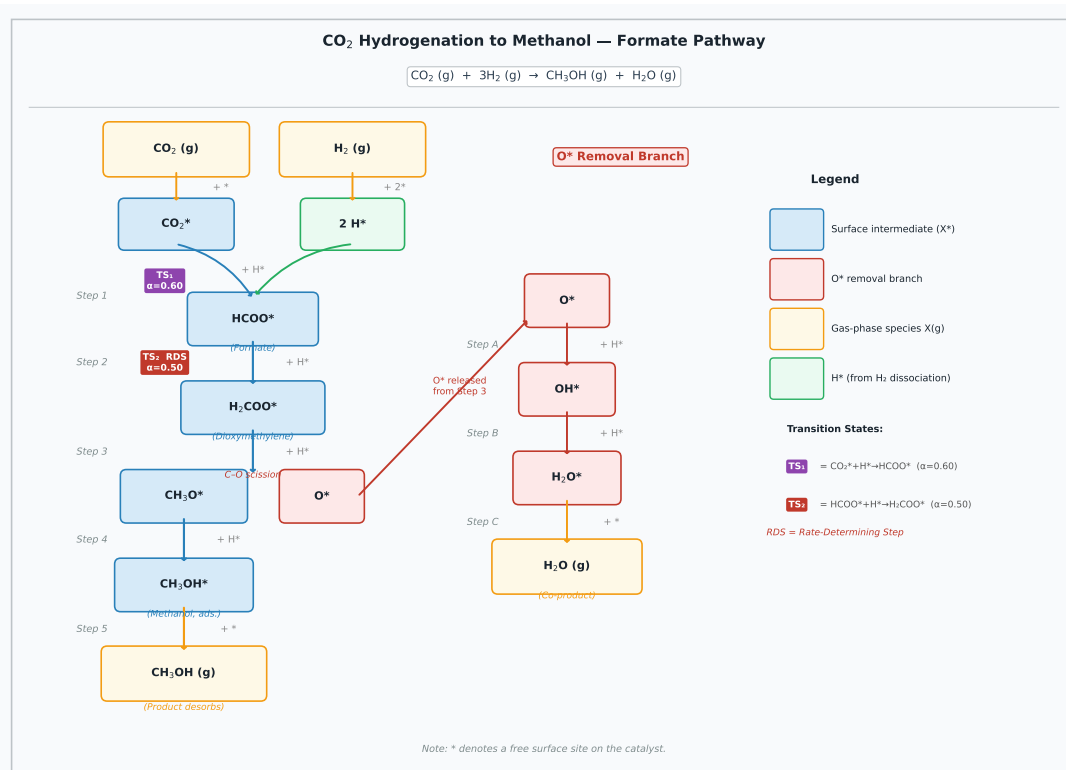


**Figure 1:** Linear scaling relations for  $\text{CH}_3\text{O}^*$  and  $\text{HCOO}^*$  adsorption energies across eight transition metals as functions of  $\text{CO}^*$  and  $\text{O}^*$  descriptors. Both relations exhibit strong linearity ( $R^2 > 0.95$ ), validating the two-descriptor framework. The stronger  $\text{O}^*$  dependence reflects oxygen-mediated binding of both intermediates.

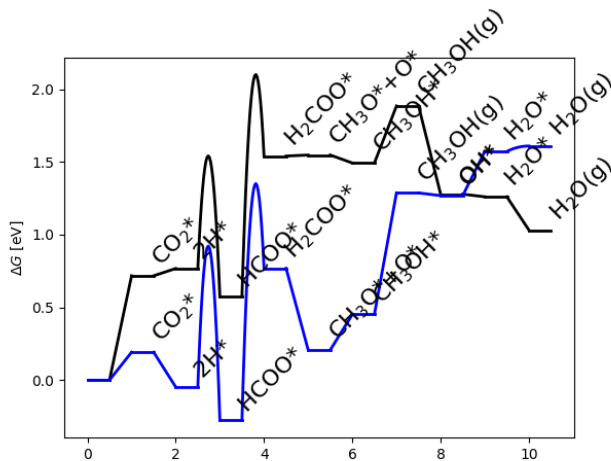
#### 3.3.1 Cu(111)

The Cu FED reveals a nearly thermoneutral  $\text{CO}_2$  adsorption step ( $\Delta G \approx -0.02 \text{ eV}$ ). The first hydrogenation ( $\text{CO}_2^* + \text{H}^* \rightarrow \text{HCOO}^*$ ) proceeds over a TS at  $G \approx +0.63 \text{ eV}$  (gas reference), corresponding to an activation free energy of  $\approx 0.80 \text{ eV}$  from the co-adsorbed reactant state in close agreement with the  $0.77 \text{ eV}$  NEB value reported by Grabow and Mavrikakis [5].

The primary kinetic bottleneck is the second hydrogenation-



**Figure 2:** Formate pathway for CO<sub>2</sub> hydrogenation to methanol on a transition metal catalyst surface. The empty site on the catalyst surface is denoted by the star (\*) symbol. TS<sub>1</sub> and TS<sub>2</sub> denote the transition states for the first and second hydrogenation steps, respectively; TS<sub>2</sub> is the rate-determining step on Cu(111).



**Figure 3:** Free energy diagrams at  $T = 513\text{ K}$  for the formate pathway on Cu(111) (black, solid) and Ni(111) (blue, solid). Energies in eV, referenced to  $\text{CO}_2(\text{g}) + 1.5\text{H}_2(\text{g}) = 0\text{ eV}$ . Transition state peaks (TS<sub>1</sub>, TS<sub>2</sub>) are shown for reactions 6 and 7.

tion,  $\text{HCOO}^* + \text{H}^* \rightarrow \text{H}_2\text{COO}^*$ , with TS at  $G \approx +1.01\text{ eV}$ , corresponding to an apparent barrier of  $\approx 1.4\text{ eV}$  from the  $\text{HCOO}^* + \text{H}^*$  state. This identifies  $\text{H}_2\text{COO}^*$  formation as the rate-determining step on Cu, consistent with the high  $\text{HCOO}^*$  surface coverage predicted by Grabow and Mavrikakis [5]. Downstream steps ( $\text{H}_2\text{COO}^* \rightarrow \text{CH}_3\text{O}^*$ ,  $\text{CH}_3\text{O}^* + \text{H}^* \rightarrow \text{CH}_3\text{OH}^*$ , desorption) are thermodynamically favourable.

### 3.3.2 Ni(111)

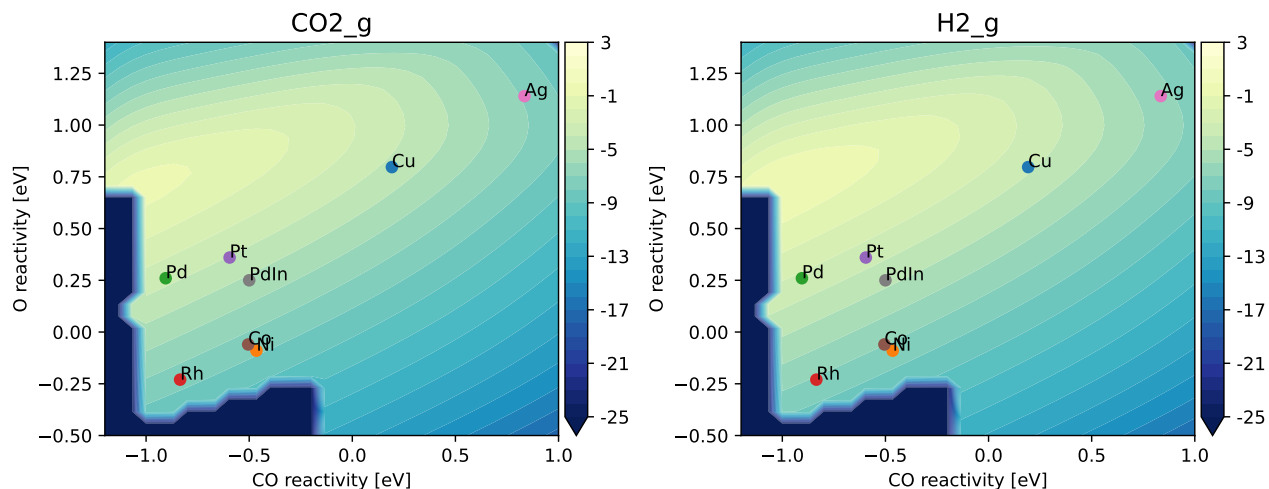
The Ni FED displays a qualitatively different character.  $\text{CO}_2^*$  is stabilised ( $E_f \approx -0.20\text{ eV}$ ), and  $\text{HCOO}^*$  occupies a deep well ( $E_f \approx -1.07\text{ eV}$ ). The pathway culminates at  $\text{CH}_3\text{O}^*$  ( $E_f \approx -1.10\text{ eV}$ ) the deepest well in the network. The deep  $\text{CH}_3\text{O}^*$  well is the primary descriptor of Ni's low methanol activity; additionally,  $\text{O}^*$  is nearly thermoneutral on Ni ( $E_f \approx -0.09\text{ eV}$ ), inhibiting water formation and causing  $\text{O}^*$ -mediated surface poisoning. These two effects explain why Ni is experimentally selective towards  $\text{CH}_4$  rather than methanol [12].

## 3.4 Volcano Plot Analysis

Figure 4 shows the two-dimensional CatMAP volcano plots for CO<sub>2</sub> (left) and H<sub>2</sub> (right) consumption TOF ( $\log_{10}(\text{TOF}), \text{s}^{-1}$ ) as a function of  $E_f(\text{CO}^*)$  and  $E_f(\text{O}^*)$  at  $T = 513\text{ K}$ .

### 3.4.1 Topology of the volcano surface

Both panels exhibit the expected asymmetric volcano shape [7, 8]. The activity ridge runs diagonally from ( $\text{CO}^* \approx -0.4\text{ eV}, \text{O}^* \approx +0.1\text{ eV}$ ) to ( $\text{CO}^* \approx +0.3\text{ eV}, \text{O}^* \approx +0.9\text{ eV}$ ), reflecting the correlated scaling of  $\text{HCOO}^*$  and  $\text{CH}_3\text{O}^*$  with both descriptors. On the weak-binding side,  $\text{CO}_2^*$  adsorption becomes endergonic, suppressing the first hydrogenation. On the strong-binding side,  $\text{HCOO}^*$  and  $\text{CH}_3\text{O}^*$  are excessively stabilised, inhibiting desorption. The unsolved white region (lower left) corresponds to near-unity co-coverages of multiple adsorbates, where the mean-field equations become ill-conditioned.



**Figure 4:** Volcanic plots for the consumption rates of (A) CO<sub>2</sub> and (B) H<sub>2</sub> over the (111) sites of monometallic transition metal catalysts at  $T = 513$  K,  $P(\text{CO}_2) = 0.2$  atm,  $P(\text{H}_2) = 0.6$  atm. Contour levels span  $\approx 25$  orders of magnitude on a  $\log_{10}$  scale. Metal data points are overlaid with labels. The dark blue region (lower left) is the strongly-binding regime where the microkinetic solver does not converge. Error bar =  $\pm 0.2$  eV.

**Table 4:** Activity ranking of the eight metals.  $\log_{10}(\text{TOF})$  values are approximate ranges read from the volcano contour map at the metal's descriptor coordinates.

| Metal | $\log_{10}(\text{TOF})$ | Zone      | Limitation                                  |
|-------|-------------------------|-----------|---------------------------------------------|
| Cu    | -5 to -3                | Near apex | $\text{HCOO}^* + \text{H}^*$ TS             |
| PdIn  | -9 to -7                | Moderate  | $\text{CO}^*$ slightly strong               |
| Pt    | -10 to -8               | Moderate  | $\text{CO}^*$ over-binding                  |
| Co    | -14 to -12              | Low       | Deep $\text{CH}_3\text{O}^*$ well           |
| Ni    | -14 to -12              | Low       | $\text{CH}_3\text{O}^* + \text{O}^*$ poison |
| Pd    | -16 to -14              | Very low  | $\text{CO}^*$ poisoning                     |
| Rh    | < -20                   | Very low  | $\text{HCOO}^* + \text{O}^*$ poison         |
| Ag    | < -20                   | Inactive  | Endergonic $\text{CO}_2$ ads.               |

### 3.4.2 Metal positions and activity ranking

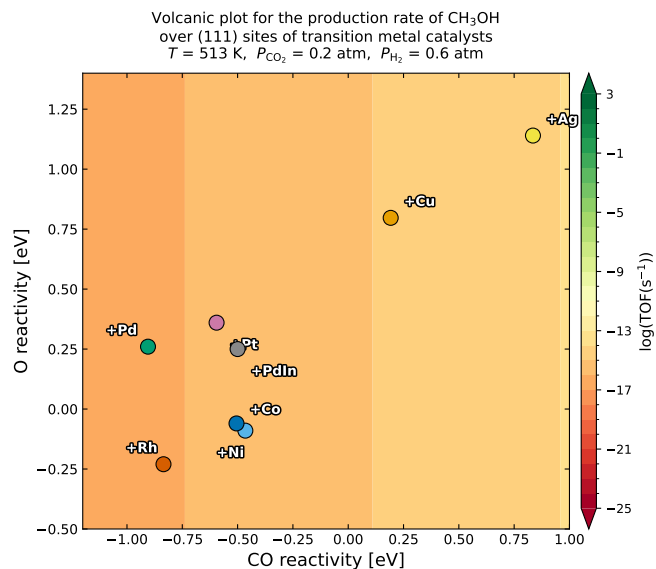
Table 4 summarises the activity ranking derived from the volcano descriptor positions.

**Cu** sits on the weak-binding shoulder of the activity ridge at  $(+0.19, +0.80)$  eV, consistent with its industrial dominance. **PdIn** at  $(-0.50, +0.25)$  eV is 4-6 orders of magnitude less active than Cu at 0.8 atm total, but the descriptor distance to the apex is only  $\approx 0.7$  eV in  $\text{CO}^*$ ; at industrial pressures (50–100 bar) this gap narrows substantially, consistent with Wu and Yang [9]. **Ni** and **Co** cluster at  $(\text{CO}^* \approx -0.47 \text{ eV}, \text{O}^* \approx -0.07 \text{ eV})$ , a zone of low activity characterised by  $\text{O}^*$  removal being uphill and deep  $\text{CH}_3\text{O}^*$  wells. **Ag** at  $(+0.84, +1.14)$  eV is completely inactive, in agreement with the absence of any Ag-based methanol synthesis catalyst in the literature.

### 3.5 Methanol Production Rate Volcano

Figure 5 shows the volcanic plot specifically for the CH<sub>3</sub>OH production rate, which directly quantifies the methanol synthesis activity of each metal as a function of descriptor energies.

The CH<sub>3</sub>OH production rate map shows the same topological features as the CO<sub>2</sub> and H<sub>2</sub> consumption maps (Figure 4), as



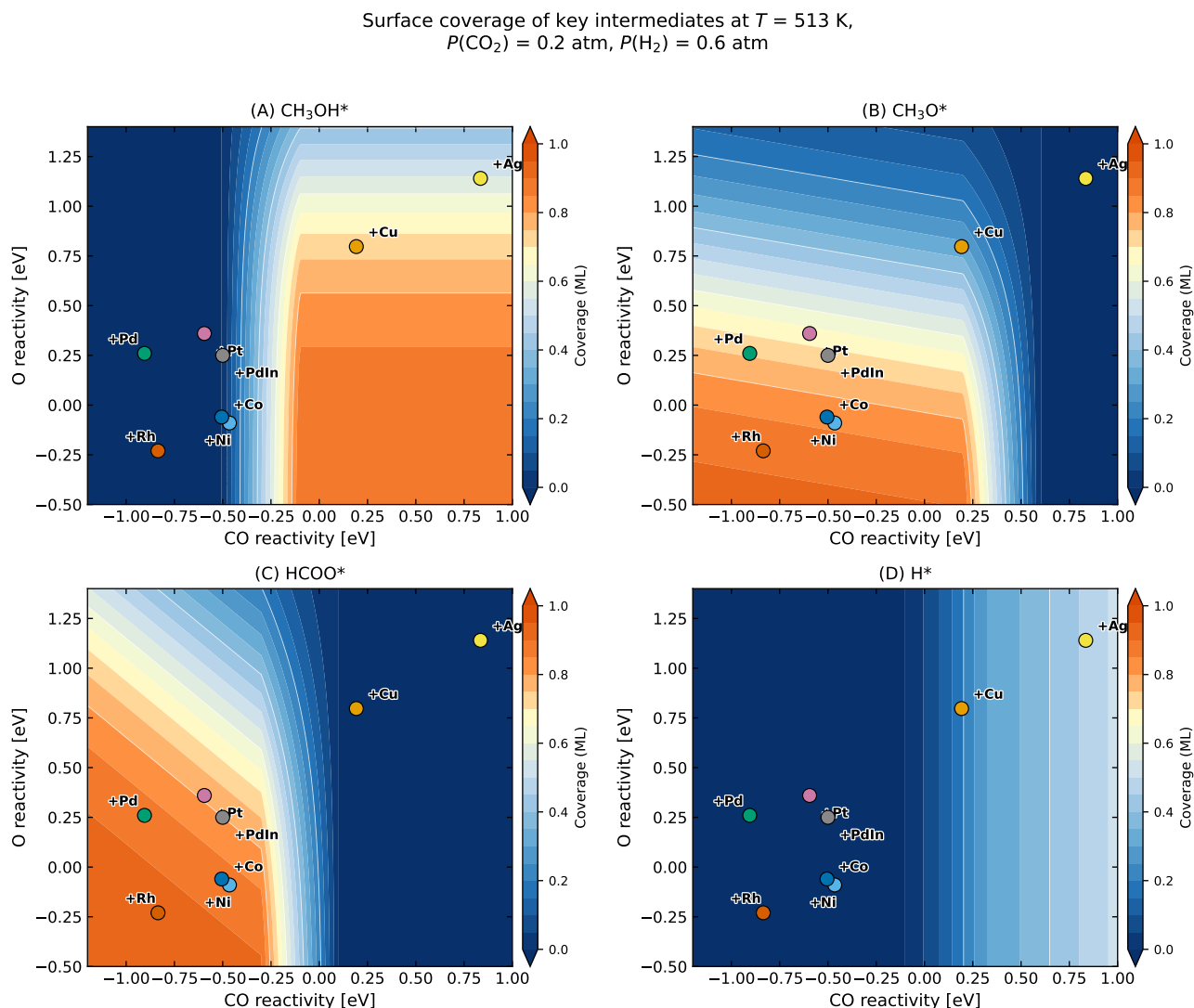
**Figure 5:** Volcanic plot for the CH<sub>3</sub>OH Formation over the (111) sites of eight transition metal catalysts at  $T = 513$  K,  $P(\text{CO}_2) = 0.2$  atm,  $P(\text{H}_2) = 0.6$  atm. The colour scale shows  $\log_{10}(\text{TOF})$  in  $\text{s}^{-1}$ . Cu sits nearest to the volcano apex, confirming its industrial dominance. Error bar =  $\pm 0.2$  eV.

expected from the fixed 1:1 stoichiometry between reactant consumption and methanol production in equation (1). Cu achieves the highest predicted CH<sub>3</sub>OH TOF among all eight metals, followed by PdIn and Pt in the intermediate activity zone.

### 3.6 Surface Coverage Analysis

Figure 6 shows the steady-state coverage maps for all 10 surface intermediates at  $T = 513$  K.

In the weak-binding zone (Cu, Ag),  $\text{CH}_3\text{OH}^*$  achieves the highest steady-state coverage (0.6–1.0 ML), indicating that



**Figure 6:** Coverage of the key surface intermediates (A)  $\text{CH}_3\text{OH}^*$ , (B)  $\text{CH}_3\text{O}^*$ , (C)  $\text{HCOO}^*$ , and (D)  $\text{H}^*$  over the (111) sites of monometallic transition metal catalysts at  $T = 513$  K,  $P(\text{CO}_2) = 0.2$  atm,  $P(\text{H}_2) = 0.6$  atm. Linear scale: 0 (dark blue) to 1 ML (orange/yellow). Metal data points are overlaid. Error bar =  $\pm 0.2$  eV.

methanol desorption is the last partially rate-limiting step at low pressure.  $\text{H}^*$  accumulates on Ag (0.6–1.0 ML) as  $\text{CO}_2$  formation is endergonic, leaving  $\text{H}^*$  as a spectator.

The most prominent feature is the large region of high  $\text{CH}_3\text{O}^*$  coverage (0.4–0.9 ML) spanning  $E_f(\text{CO}^*) \in [-1.1, +0.1]$  eV, encompassing Ni, Co, Pt, PdIn, and Pd. The deep  $\text{CH}_3\text{O}^*$  wells on these metals ( $E_f = -0.84$  to  $-1.22$  eV) create thermodynamic sinks that block active sites and suppress  $\text{CO}_2$  adsorption, directly rationalising the low predicted TOF in this descriptor region. This is corroborated by temperature-programmed surface reaction (TPSR) studies on Pd(111) and Ni(111) [12].

In the strongly-binding zone (Rh, lower Pd),  $\text{HCOO}^*$  reaches 0.4–1.0 ML. Critically,  $\text{CO}_2^*$ ,  $\text{H}_2\text{COO}^*$ ,  $\text{O}^*$ , and  $\text{OH}^*$  all show near-zero coverage ( $\ll 0.1$  ML) throughout, confirming that these are transient intermediates rather than surface-poisoning species. The formate pathway kinetics are therefore governed

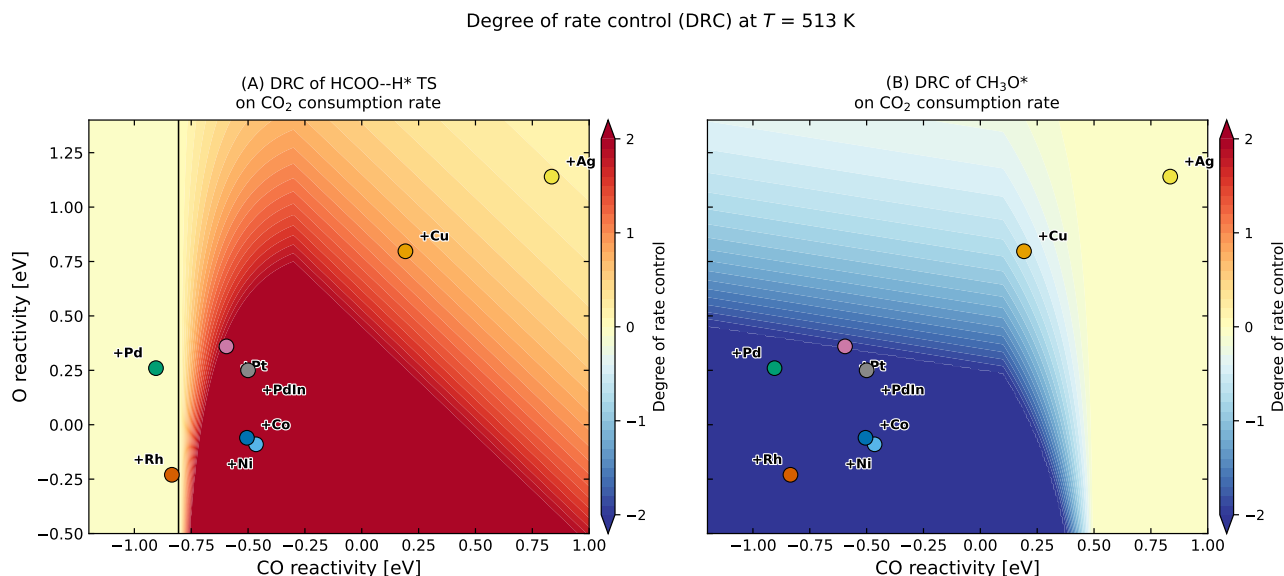
by a two-MASI model: either  $\text{CH}_3\text{O}^*$  or  $\text{HCOO}^*$  dominates depending on the descriptor region.

### 3.7 Degree of Rate Control Analysis

Figure 7 shows the DRC maps,  $X_{\text{RC},i} = (\partial \ln r / \partial \ln k_i)|_{K_{\text{eq}},k_{j \neq i}}$  [19], across the descriptor space.

In the  $\text{CO}_2\text{-g}|\text{HCOO-H-s}$  panel, the Cu descriptor zone shows  $X_{\text{RC}} \approx +1$ , directly identifying the  $\text{HCOO}^* + \text{H}^* \rightarrow \text{H}_2\text{COO}^*$  step as rate-limiting for  $\text{CO}_2$  consumption on Cu. A secondary positive contribution ( $X_{\text{RC}} \approx +0.5$ ) appears in the  $\text{CO}_2\text{-g}|\text{CO}_2\text{-H-s}$  panel, indicating that the  $\text{CO}_2^* + \text{H}^* \rightarrow \text{HCOO}^*$  step is partially co-limiting. These results are in excellent quantitative agreement with the DRC analysis of Grabow and Mavrikakis [5].

Negative DRC values quantify surface inhibition. The  $\text{CO}_2\text{-g}|\text{CH}_3\text{O-s}$  panel shows  $X_{\text{RC}} \approx -1$  to  $-2$  across the Ni/Co/Pt/PdIn zone, confirming  $\text{CH}_3\text{O}^*$  as the primary inhibiting MASI. The  $\text{CO}_2\text{-g}|\text{HCOO-s}$  panel shows  $X_{\text{RC}} \approx -1$



**Figure 7:** Degree of rate control (DRC) of (A) the  $\text{HCOO-H}^*$  transition state ( $\text{TS}_2$ ) and (B) the  $\text{CH}_3\text{O}^*$  intermediate on the  $\text{CO}_2$  consumption rate, over the (111) sites of monometallic transition metal catalysts at  $T = 513$  K. Scale:  $-2$  (blue, surface inhibitor/MASI) to  $+2$  (yellow, rate-limiting step). Error bar  $= \pm 0.2$  eV.

in the Rh/Pd region, confirming  $\text{HCOO}^*$  poisoning. The spatial boundary between positive  $\text{HCOO-H}$  DRC and negative  $\text{CH}_3\text{O}$  DRC precisely coincides with the volcano activity ridge, providing a mechanistic explanation for the volcano shape.

The  $\text{H}_2\text{g}$  panels are essentially identical to the  $\text{CO}_2\text{g}$  panels, as expected from the fixed 1:3  $\text{CO}_2\text{:H}_2$  stoichiometry. Panels involving  $\text{CO}_\text{g}$  reflect numerical noise from the dummy adsorption step and are disregarded.

### 3.8 Catalyst Design Principles

Three quantitative design principles emerge:

**1. Target**  $E_f(\text{CO}^*) \in [-0.2, +0.2]$  eV and  $E_f(\text{O}^*) \in [+0.5, +1.0]$  eV. This window defines the volcano apex where Cu resides. Interventions that shift  $\text{CO}^*$  by  $-0.1$  to  $-0.3$  eV relative to Cu while preserving  $E_f(\text{O}^*) > +0.5$  eV should improve activity. ZnO promotion of Cu achieves this by stabilising formate at the Cu-Zn interface [5].

**2. Exploit intermetallic alloying** to shift strongly-binding metals toward the apex. The shift from Pd ( $E_f(\text{CO}^*) = -0.91$  eV) to PdIn ( $-0.50$  eV) is  $+0.41$  eV along  $\text{CO}^*$  a movement that translates to 4-6 orders of magnitude improvement in low-pressure TOF. Similar electronic modification with Ga [17] or Zn [21] follows the same principle.

**3.  $\text{O}^*$  removal is a co-limiting constraint.** Metals with  $E_f(\text{O}^*) < 0$  (Ni, Co, Rh) are deactivated regardless of their  $\text{CO}^*$  position because water formation is uphill. Catalyst design must simultaneously target both descriptors. This dual constraint explains why the optimal window is narrow and why significant improvements over Cu require breaking the transition metal scaling line for example, through bifunctional catalysts where  $\text{O}^*$  removal occurs on a separate oxide site [20].

## 4. Conclusions

A first-principles-rooted microkinetic model of the  $\text{CO}_2 \rightarrow \text{CH}_3\text{OH}$  formate pathway has been constructed and solved across a  $20 \times 20$  descriptor grid using CatMAP [1] for eight transition metal catalysts. The principal conclusions are:

1. Cu(111) sits near the volcano apex at  $(E_f(\text{CO}^*), E_f(\text{O}^*)) = (+0.19, +0.80)$  eV, consistent with its industrial dominance. The primary kinetic limitation is the  $\text{HCOO}^* + \text{H}^* \rightarrow \text{H}_2\text{COO}^*$  step ( $\approx 1.4$  eV apparent barrier), in agreement with Grabow and Mavrikakis [5].
2. PdIn shows moderate activity at  $(-0.50, +0.25)$  eV, representing a significant improvement over monometallic Pd via  $\text{CO}^*$  weakening by In, consistent with experimental methanol selectivity data [9].
3. Ni and Co are deactivated by deep  $\text{CH}_3\text{O}^*$  wells ( $E_f \approx -1.10$  eV) and near-zero  $\text{O}^*$  energies, explaining their experimental selectivity towards methanation products [12].
4. BEP scaling relations for  $\text{HCOO}^*$  and  $\text{CH}_3\text{O}^*$  versus  $\text{CO}^*$  exhibit  $R^2 > 0.95$ , validating the two-descriptor framework [6, 7].
5. Rational catalyst design requires simultaneously targeting  $E_f(\text{CO}^*) \in [-0.2, +0.2]$  eV and  $E_f(\text{O}^*) \in [+0.5, +1.0]$  eV a window most readily accessed through intermetallic or bimetallic strategies [17, 21].

Future work should extend the model to industrial pressures (50–100 bar), incorporate adsorbate-adsorbate interactions, and include the competing RWGS pathway for completeness.

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